

PRACTICAL DBMS

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COURSE CODE: CSA0503

COURSE NAME: DATABASE MANAGEMENT SYSTEMS

EXPERIMENT NO:1 DDL Commands – CREATE, ALTER, DROP

AIM:

To Create, Alter, Drop, Rename, Truncate using Data Definition Language (DDL) statements.

Description:

Data Definition Language (DDL) statements are used to define the database structure or schema.

DDL Commands: Create, Alter, Drop, Rename, Truncate

- CREATE - to create objects in the database
- ALTER - alters the structure of the database
- DROP - delete objects from the database
- TRUNCATE - remove all records from a table, including all spaces allocated for the records are removed
- RENAME - rename an object

SYNTAX:

CREATE TABLE

CREATE TABLE table_name

```
(  
column_name1 data_type,  
column_name2 data_type,  
column_name3 data_type,  
....  
);
```

```
mysql> create table STUDENT(RegNo int,Name varchar(15),Gender v
archar(1),DOB date,MobileNo int,City varchar(15));
Query OK, 0 rows affected (0.37 sec)
```

```
mysql> desc STUDENT;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	YES		NULL	
Name	varchar(15)	YES		NULL	
Gender	varchar(1)	YES		NULL	
DOB	date	YES		NULL	
MobileNo	int	YES		NULL	
City	varchar(15)	YES		NULL	

```
6 rows in set (0.12 sec)
```

```
mysql> create table FACULTY(FacNo varchar(4),FacName varchar(15)
,Gender varchar(1),DOB date,DOJ date,MobileNo int);
Query OK, 0 rows affected (0.14 sec)
```

```
mysql> desc FACULTY;
```

Field	Type	Null	Key	Default	Extra
FacNo	varchar(4)	YES		NULL	
FacName	varchar(15)	YES		NULL	
Gender	varchar(1)	YES		NULL	
DOB	date	YES		NULL	
DOJ	date	YES		NULL	
MobileNo	int	YES		NULL	

```
6 rows in set (0.01 sec)
```

```
mysql> create table DEPARTMENT(DeptNo varchar(4),DeptName varcha
r(15),DeptHead varchar(4));
Query OK, 0 rows affected (0.14 sec)
```

```
mysql> desc DEPARTMENT;
```

Field	Type	Null	Key	Default	Extra
DeptNo	varchar(4)	YES		NULL	
DeptName	varchar(15)	YES		NULL	
DeptHead	varchar(4)	YES		NULL	

```
3 rows in set (0.03 sec)
```

```
mysql> create table COURSE(CourseNo varchar(3),CourseDesc varchar(14),CourseType varchar(1),SemNo varchar(1),HallNo varchar(4),FacNo varchar(4));
Query OK, 0 rows affected (0.12 sec)
```

```
mysql> desc COURSE;
```

Field	Type	Null	Key	Default	Extra
CourseNo	varchar(3)	YES		NULL	
CourseDesc	varchar(14)	YES		NULL	
CourseType	varchar(1)	YES		NULL	
SemNo	varchar(1)	YES		NULL	
HallNo	varchar(4)	YES		NULL	
FacNo	varchar(4)	YES		NULL	

6 rows in set (0.01 sec)

ALTER A TABLE

To add a column in a table

```
ALTER TABLE table_name
```

```
ADD column_namedatatype;
```

To delete a column in a table

```
ALTER TABLE table_name
```

```
DROP COLUMN column_name;
```

```
mysql> alter table faculty add Dept varchar(10);
Query OK, 0 rows affected (0.27 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Right syntax to use near () at line 2

```
mysql> desc FACULTY;
```

Field	Type	Null	Key	Default	Extra
FacNo	varchar(4)	YES		NULL	
FacName	varchar(15)	YES		NULL	
Gender	varchar(1)	YES		NULL	
DOB	date	YES		NULL	
DOJ	date	YES		NULL	
MobileNo	int	YES		NULL	
Dept	varchar(10)	YES		NULL	

DROP TABLE

DROP TABLE table_name;

```
mysql> DROP TABLE student;
Query OK, 0 rows affected (0.08 sec)

mysql> desc student;
ERROR 1146 (42S02): Table 'avanthika.student' doesn't exist
mysql> |
```

TRUNCATE TABLE

TRUNCATE TABLE table_name;

```
mysql> truncate table student;  
Query OK, 0 rows affected (0.09 sec)
```

```
mysql> desc student;
```

Field	Type	Null	Key	Default	Extra
regno	int	YES		NULL	
name	varchar(15)	YES		NULL	
gender	varchar(1)	YES		NULL	
DOB	date	YES		NULL	
mobilenno	int	YES		NULL	
city	varchar(15)	YES		NULL	

6 rows in set (0.01 sec)

RESULT:

Tables are created, altered and modified using DDL commands.

EXPERIMENT NO:2 DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK

AIM:

To add the constraints like primary key, foreign key, unique key and check using DDL commands.

Description:

PRIMARY KEY:

The PRIMARY KEY constraint uniquely identifies each record in a database table.

Primary keys must contain UNIQUE values, and cannot contain NULL values.

A table can have only one primary key, which may consist of single or multiple fields.

FOREIGN KEY:

➤ A FOREIGN KEY is a key used to link two tables together.

➤ A FOREIGN KEY is a field (or collection of fields) in one table that refers to the PRIMARY KEY in another table.

➤ The table containing the foreign key is called the child table, and the table containing the candidate key is

called the referenced or parent table.

UNIQUE Constraint:

➤ The UNIQUE constraint ensures that all values in a column are different.

➤ Both the UNIQUE and PRIMARY KEY constraints provide a guarantee for uniqueness for a column

or set of columns.

➤ A PRIMARY KEY constraint automatically has a UNIQUE constraint.

➤ However, you can have many UNIQUE constraints per table, but only one PRIMARY KEY

constraint per table.

CHECK Constraint:

➤ The CHECK constraint is used to limit the value range that can be placed in a column

➤ If you define a CHECK constraint on a single column it allows only certain values for this column.

➤ If you define a CHECK constraint on a table it can limit the values in certain columns based on

values in other columns in the row.

PRIMARY:

ALTER TABLE table_name

ADD PRIMARY KEY(primary_key_column);

```
mysql> alter table student add primary key(regno);
Query OK, 0 rows affected (0.18 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc student;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| regno | int           | NO   | PRI | NULL    |       |
| gender | varchar(15)   | YES  |     | NULL    |       |
| name   | varchar(15)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

FOREIGN KEY:

ALTER TABLE table_name

ADD CONSTRAINT constraint_name

FOREIGN KEY foreign_key_name (columns)

REFERENCES parent_table(columns)

ON DELETE action

ON UPDATE action

UNIQUE:

CREATE TABLE table_1(

...

column_name_1 data_type,

...

UNIQUE(column_name_1)

```
);
CHECK
CREATE TABLE IF NOT EXISTS parts (
part_no VARCHAR(18) PRIMARY KEY,
description VARCHAR(40),
cost DECIMAL(10 , 2 ) NOT NULL CHECK(cost > 0), price DECIMAL (10,2) NOT
NULL
);
```

```
mysql> alter table student add unique key(name);
Query OK, 0 rows affected (0.15 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc student;
```

Field	Type	Null	Key	Default	Extra
regno	int	NO	PRI	NULL	
gender	varchar(15)	YES		NULL	
name	varchar(15)	YES	UNI	NULL	

3 rows in set (0.01 sec)

Result:

DDL Commands with Primary, Foreign, Unique, Check constraints are updated and verified.

EXPERIMENT NO:3 DML Commands – INSERT, SELECT

AIM:

To perform Data Manipulation Language (DML) Commands such as INSERT, SELECT in the table.

Description:

Data Manipulation Language (DML) statements are used for managing data within schema objects. DML

Commands: Insert, Select

➤ INSERT - insert data into a table

➤ SELECT - retrieve data from the a database

INSERT:

INSERT INTO table_name

VALUES (value1, value2, value3,...);

(or)

INSERT INTO table_name (column1, column2, column3,...)

VALUES (value1, value2, value3,...);

SELECT:

SELECT column_name(s)

FROM table_name;

```
mysql> insert into student values(192511149,'f','devi'),(192521245,'f','harini'),(192521137,'m','jagan');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> select * from student;
+-----+-----+-----+
| regno | gender | name  |
+-----+-----+-----+
| 192511149 | f      | devi  |
| 192521137 | m      | jagan |
| 192521245 | f      | harini |
+-----+-----+-----+
3 rows in set (0.01 sec)
```

```
mysql>
```

RESULT:

Data Manipulation Language (DML) Commands such as INSERT, SELECT are performed in the five tables.

EXPERIMENT NO: 4 DML Commands with Constraints –UPDATE, DELETE

AIM:

To perform Data Manipulation Language (DML) Commands such as UPDATE, DELETE in the table.

Description:

Data Manipulation Language (DML) statements are used for managing data within schema objects. DML

Commands: Update, Delete

➤ UPDATE - updates existing data within a table

➤ DELETE - deletes all records from a table, the space for the records remain

UPDATE:

UPDATE table_name

SET column1=value, column2=value2,...

WHERE some_column=some_value;

```
mysql> update student set name='swetha' where regno=192511149;
Query OK, 1 row affected (0.10 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> select * from student;
+-----+-----+-----+
| regno   | gender | name   |
+-----+-----+-----+
| 192511149 | f      | swetha |
| 192521137 | m      | jagan  |
| 192521245 | f      | harini |
+-----+-----+-----+
3 rows in set (0.02 sec)
```

DELETE:

DELETE FROM table_name

WHERE some_column=some_value;

```
mysql> delete from student where name='jagan';  
Query OK, 1 row affected (0.03 sec)
```

```
mysql> select * from student;  
+-----+-----+-----+  
| regno   | gender | name   |  
+-----+-----+-----+  
| 192511149 | f      | swetha |  
| 192521245 | f      | harini |  
+-----+-----+-----+  
2 rows in set (0.00 sec)
```

RESULT:

Data Manipulation Language (DML) Commands such as UPDATE, DELETE are performed in the five tables.

EXPERIMENT NO:5 SELECT with various clause – WHERE, pattern matching

AIM:

To view the records from the tables using SELECT commands with WHERE Clause and Pattern matching.

DESCRIPTION:

The SELECT statement allows you to get the data from tables. A table consists of rows and columns like a

spreadsheet. Often, you want to see a subset rows, a subset of columns, or a combination of two. The result of the

SELECT statement is called a result set that is a list of rows, each consisting of the same number of columns.

SELECT:

SELECT column_1, column_2, ...

FROM table_1

[INNER | LEFT | RIGHT] JOIN table_2 ON conditions

WHERE conditions

GROUP BY column_1

HAVING group_conditions

ORDER BY column_1

LIMIT offset, length;

The SELECT statement consists of several clauses as explained in the following list:

- SELECT followed by a list of comma-separated columns or an asterisk (*) to indicate that you want to return all columns.
- FROM specifies the table or view where you want to query the data.
- JOIN gets related data from other tables based on specific join conditions.
- WHERE clause filters row in the result set.
- GROUP BY clause groups a set of rows into groups and applies aggregate functions on each group.
- HAVING clause filters group based on groups defined by GROUP BY clause.
- ORDER BY clause specifies a list of columns for sorting.

➤ LIMIT constrains the number of returned rows.

```
mysql> select name,gender from student;
+-----+-----+
| name  | gender |
+-----+-----+
| swetha | f      |
| harini | f      |
+-----+-----+
2 rows in set (0.10 sec)
```

```
mysql> select * from student;
+-----+-----+-----+
| regno  | gender | name  |
+-----+-----+-----+
| 192435163 | m      | manoj |
| 192453678 | m      | sanoop |
| 192511149 | f      | swetha |
| 192521245 | f      | harini |
+-----+-----+-----+
4 rows in set (0.01 sec)
```

```
mysql> select * from student where gender='f';
+-----+-----+-----+
| regno  | gender | name  |
+-----+-----+-----+
| 192511149 | f      | swetha |
| 192521245 | f      | harini |
+-----+-----+-----+
2 rows in set (0.01 sec)
```

LIKE:

➤ The LIKE operator is commonly used to select data based on patterns. Using the LIKE operator in the right

way is essential to increase the query performance.

➤ The LIKE operator allows you to select data from a table based on a specified pattern. Therefore, the LIKE

operator is often used in the WHERE clause of the SELECT statement.

➤ MySQL provides two wildcard characters for using with the LIKE operator, the percentage % and

underscore _ .

➤ The percentage (%) wildcard allows you to match any string of zero or more characters.

➤ The underscore (_) wildcard allows you to match any single character.

```
mysql> select * from student where name like 's%';
+-----+-----+-----+
| regno   | gender | name   |
+-----+-----+-----+
| 192453678 | m     | sanoop |
| 192511149 | f     | swetha |
+-----+-----+-----+
2 rows in set (0.02 sec)
```

RESULT:

The records from the tables are displayed using SELECT commands with WHERE Clause and Pattern matching.